

(Information Security system)

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• Section - A (2x2=4)

1. Define cryptography?

Ans Cryptography is the practice of securing information by converting it into an unreadable format (encryption) to protect it from unauthorized access, ensuring confidentiality, integrity, authentication and non-repudiation.

2. Define steganography?

Ans Steganography is the technique of hiding secret information within non-secret data, such as images, audio, or text to conceal its existence and prevent detection by unauthorized parties.

• Section - B (2x7=14)

1. Define symmetric and asymmetric key?

Ans Cryptographic system use keys to encrypt and decrypt data. These systems are classified into symmetric and asymmetric key cryptography.

(i) Symmetric key

- Uses a single key for both encryption & decryption.
- Faster and more efficient for large data transmission.
- Requires secure key exchanges between sender and receiver.

Ex: AES (Advanced Encryption Standard)

- DES (Data Encryption Standard)

Real-world use: Secure File storage, VPN Encryption.

(ii) Asymmetric key :-

- Use two key - public key for encryption
private key for decryption

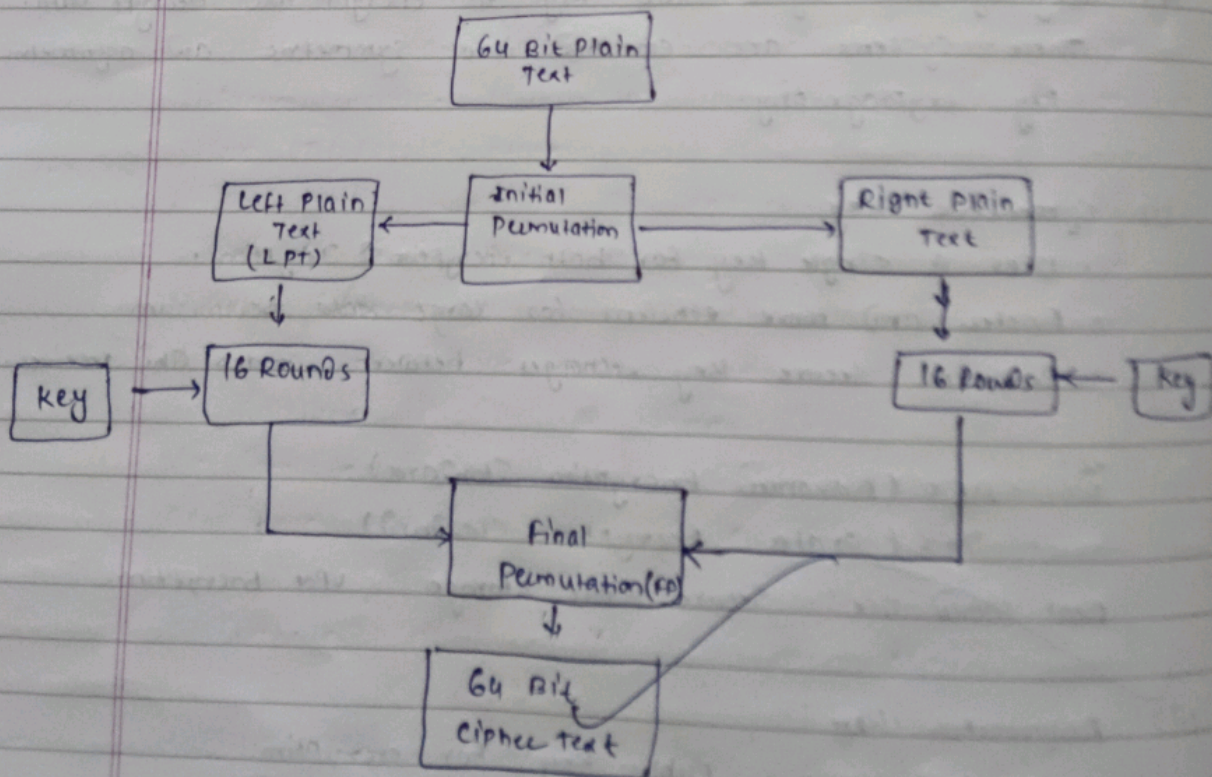
- Slower but more secure due to key pair usage.
- Solve the key exchange problem of symmetric encryption.

ex: • RSA (Rivest-Shamir-Adleman)
• ECC (Elliptic Curve Cryptography)

Real world use:- Digital signatures, SSL/TLS for secure web browsing.

2. Explain DES and security of DES?

Ans Data Encryption Standard (DES) is a symmetric key encryption algorithm developed by IBM in the 1970s and later adopted as a U.S. government standard. It uses a 56-bit key and operates on 64-bit data blocks through a series of complex transformations.



• How DES Works:-

- (i) Initial permutation (IP):- Rearrange input bits.
- (ii) Key Scheduling - Generates 16 subkeys from the 56-bit key.
- (iii) Feistel Function (F) - Applies substitution and permutation.
- (iv) XOR Operation - Mixes data with key-based transformations.
- (v) Final permutation (FP) - Produces the encrypted output.

• Security of DES:-

- Key Length - DES uses a 56-bit key, which provides 2^{56} possible keys. This was considered secure in the 1970s, but is now vulnerable to brute-force attacks due to advances in computing power.
- Cryptanalytic Attacks - Techniques like differential cryptanalysis and linear cryptanalysis can exploit weaknesses in DES, though it was designed to resist these attacks.
- Enhancements - To address its weakness, Triple DES (3DES) was introduced, which applies DES three times with two or three keys, increasing effective key length to 112 or 168 bits.

• Section - C (2x11=22)

(1)(a) Explain principle of public key cryptosystem?

Ans The principle of a public key cryptosystem is based on asymmetric encryption, which uses two keys:-

- (i) Public key - Available to everyone and used for encryption.
- (ii) Private key - Kept secret and used for decryption.

Working-principle:-

- (i) The sender encrypts the message using the recipient's public key.
- (ii) Only the recipient can decrypt it using their private key.

- (iii) This ensures Confidentiality since only the intended recipient can read the message.
- (iv) It also enables authentication and integrity using digital signatures, where a sender signs a message with their private key, and the recipient verifies it using the sender's public key.

Q: RSA (Rivest - Shamir - Adleman) is a widely used public key cryptosystem.

Advantages:-

- Eliminates key distribution problems of symmetric encryption.
- Enables secure communication over untrusted networks.

(1) (b) What is RSA Algorithm and explain analysis of RSA?

Ans: RSA (Rivest - Shamir - Adleman) is a widely used public-key cryptosystem based on the difficulty of factoring large prime numbers. It is used for secure data transmission and digital signatures.

• Steps of RSA Algorithm:-

(i) Key Generation:-

- Select two large prime numbers, p and q .
- Compute $n = p \times q$ (modulus).
- Compute $\phi(n) = (p-1) \times (q-1)$.
- Choose e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
- Compute d as $d \equiv e^{-1} \pmod{\phi(n)}$.

• Public Key: (e, n) & Private Key: (d, n) .

(ii) Encryption:

$$\text{Ciphertext: } C = M^e \text{ mod } n.$$

(iii) Decryption:

$$\text{Plain text: } M = C^d \text{ mod } n.$$

• Analysis of RSA:-

- Security - Based on the difficulty of factoring large numbers.
- Key Size - Larger key sizes (e.g. 2048 bit) provide stronger security.
- Performance - Slower than symmetric encryption but ensure better security.
- Vulnerabilities:- Susceptible to brute-force attacks if weak keys are used.

2 (a) What is Diffie-Hellman key exchange algorithm?

Ans The Diffie-Hellman (DH) algorithm is a key exchange protocol that allows two parties to securely share a secret key over an insecure channel without transmitting the key directly.

• Steps of the algorithm:-

(i) public Parameters:-

- choose a large prime number p and a primitive root g (both are public).

(ii) Key Exchange -

- User A selects a private key a and computes $A = g^a \text{ mod } p$.
- User B selects a private key b and computes $B = g^b \text{ mod } p$.
- Both exchange A and B publicly.

(iii) Shared Secret Computation:-

- User A computes $K = B^a \text{ mod } p$.

- User B computes $K = A^b \text{ mod } P$.
- Since $B^a = A^b$, both get the same secret key K .

Let see the example:

$g = 7, P = 23$

A = $a = 3$ B = $b = 6$

$A = g^a \text{ mod } P$ $B = g^b \text{ mod } P$

$A = (7)^3 \text{ mod } 23$ $B = (7)^6 \text{ mod } 23$

$= 20$ $= 4$

$K = B^a \text{ mod } P$ $K = A^b \text{ mod } P$

$= (4)^3 \text{ mod } 23$ $= (20)^6 \text{ mod } 23$

$K = 18$ $= 18$

shared key

(2) (b) What is AES structure and explain Analysis of AES?

Ans AES (Advanced ~~symmetric~~ Encryption Standard) is a symmetric key encryption algorithm that encrypts data in fixed 128-bit blocks using 128, 192, or 256-bit keys. It is based on the Substitution-Permutation Network (SPN) principle.

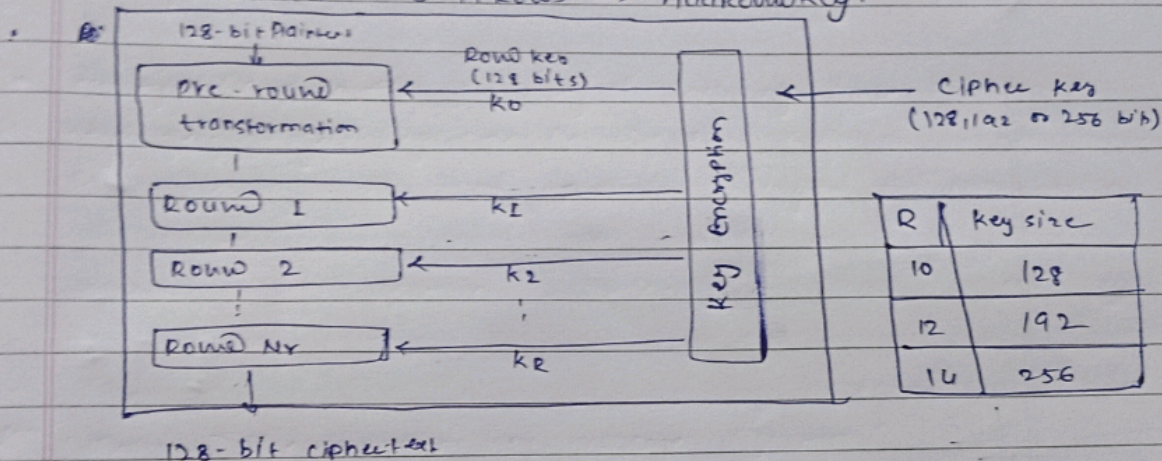
• AES structure (for 128-bit key):-

- Key Expansion - Generates round keys from the main key.
- Initial Round -
 - AddRoundKey - XOR plaintext with the first round key.
- Main rounds (9 rounds for 128-bit key):-
 - SubByte - Byte substitution using an S-box.

- ShiftRows - Left circular shift of rows.
- MixColumns - Matrix multiplication in $GF(2^8)$.
- AddRoundKey - XOR with round key.

(iv) Final Round

• SubBytes $\rightarrow$ ShiftRows $\rightarrow$ AddRoundKey.



• Analysis of AES :-

- Security - Resistant to known attack (brute force, differential, and linear cryptanalysis).
- Performance - Faster than DES and RSA, efficient for software and hardware.
- Key Size Impact - Larger key (192 or 256 bits) enhance security but increase processing time.
- Vulnerabilities - Side-channel attack (timing, power analysis) if not implemented securely.

Handwritten signature
25/8/18